Symbols used in 'dirac16complex: a second-quantized Pin(4,4) fermion field and the 6-planes it creates'

symbol	meaning	definition
$x_0, \ldots, x_7$	coordinates	x_0 hidden, $x_{1,2,3}$ space, x_4 time, $x_{5,6,7}$ superluminal time
η_{AB}	metric	$\mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$
σ	8x8 spinor metric	$\begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$
τ_A, $\bar{\tau}_A$	8x8 building blocks	$\bar{\tau}_A = \sigma \tau_A^T \sigma$, $\frac{1}{2}(\tau_A \bar{\tau}_B + \tau_B \bar{\tau}_A) = \eta_{AB} I_8$
$T^A = (T16^A)[A]$	real 16x16 Dirac matrices	$\begin{pmatrix} 0 & \bar{\tau}_A \\ \tau_A & 0 \end{pmatrix}$, $\{T^A, T^B\} = 2\eta^{AB} I_{16}$
Γ	chirality	$T^0 T^1 \cdots T^7 = \mathrm{diag}(-I_8, I_8)$
σ_{16}	Dirac-adjoint matrix	$T^0 T^1 T^2 T^3 = \mathrm{diag}(-\sigma, \sigma)$
S^{AB}	Spin(4,4) generators	$\frac{1}{4}[T^A, T^B]$
ψ = dirac16complex	16-component complex Grassmann spinor field (second quantized)	$\psi_a = \frac{1}{\sqrt{2}}(\xi_a + i\chi_a)$, $\psi_a \psi_b = -\psi_b \psi_a$
ξ_a, χ_a	real Grassmann generators	$\xi_a \xi_b = -\xi_b \xi_a$
ψ^*, $\psi^\dagger$	conjugate, conjugate transpose	$(\theta\eta)^* = \eta^* \theta^*$
$\bar{\psi}$	Dirac adjoint	$\psi^\dagger \sigma_{16}$
$\mathcal{L}$	Lagrangian density	$\frac{1}{2}(\bar{\psi} T^A \partial_A \psi - (\partial_A \bar{\psi}) T^A \psi) - m\bar{\psi}\psi - \frac{\lambda}{2}(\bar{\psi}\psi)^2$
m	mass parameter	real constant; $m = -2M$ relative to Pre-gravity6plane.nb
λ	quartic self-coupling	real constant
M, H	mass and Hubble parameters of Pre-gravity6plane.nb	$L_a = \frac{1}{H}\Psi_{16}^T \sigma_{16} \sum T^A \partial_A \Psi_{16} + \frac{2M}{H}\Psi_{16}^T \sigma_{16} \Psi_{16}$
$\partial^L / \partial \psi_a^*$	left Grassmann derivative	removes ψ_a^* from the left, sign $(-1)^{\mathrm{position}-1}$
J^A	U(1) Noether current	$\bar{\psi} T^A \psi$, $\partial_A J^A = 0$
π	canonical momentum	$\bar{\psi} T^4$
K	Krein operator	$-i\sigma_{16} T^4$, $K^\dagger = K$, $K^2 = I_{16}$
β	Hermitian involution	$-iT^4$; $\sigma_{16} = K\beta$
g_{45}	rotation by π in the (x_4, x_5)-plane	$T^4 T^5$
$\psi_\pm$	mirror sectors	$K\psi_\pm = \pm\psi_\pm$, $\psi = \psi_+ + \psi_-$
$\mathcal{L}_D$	8-component Dirac Lagrangian	$\chi^\dagger \beta (T^A \partial_A - m)\chi$, $A = 0..4$
$S_\pm$	sector scalars	$\psi_\pm^\dagger \beta \psi_\pm$; $\bar{\psi}\psi = S_+ - S_-$
Ψ_{16}, $f16[k]$	Universe wave function of Pre-gravity6plane.nb	$\Psi_{16} = (f16[0][x_0, x_4], \ldots, f16[15][x_0, x_4])$
z, t	scaled variables	$z = 6Hx_0$, $t = Hx_4$
p, q, C	exponents of light-cone modes	$\psi = e^{pz+qt}u$, $(6p)^2 - q^2 = m^2/H^2$, $6p + q = 2C$
u, w	constant spinors	$u = (6pT^0 + qT^4 - \frac{2M}{H})w$; plane wave $u = (ikT^0 - iET^4 + m)w$
k, E	wave number, frequency	$E = \pm\sqrt{k^2 + m^2}$
z_{adv}, z_{ret}, ξ_{adv}, ξ_{ret}	light-cone variables	$6t \pm z$; $x_0 \mp x_4$
u (unit spinor)	triality unit	$\frac{1}{\sqrt{2}}(1,0,0,0,1,0,0,0)$, $u^T \sigma u = 1$
${F_A}^a$, ${F_a}^A$	triality frames	$({F_A}^a)_{aA} = \eta_{AA}(\tau_A u)_a$, $({F_a}^A)_{Aa} = (u^T \sigma \bar{\tau}_A)_a$
ψ_1, ψ_2	type-1, type-2 halves	components 0..7 and 8..15
V_1, V_2	vector forms	$V_1 = \eta(\psi_1^T \sigma {F_A}^a)$, $V_2 = {F_a}^A \psi_2$
n_1, n_2	real normals	$n_k = V_k[\mathrm{Re}\,\psi]$
Z, z_i	point of $\mathbb{R}^8$	$Z = (z_1, \ldots, z_8)$
7-plane	hyperplane	$\{Z : n \cdot Z = 0\}$
M^6	6-plane	$\{Z : n_1 \cdot Z = 0,\ n_2 \cdot Z = 0\}$; 28 parameterizations
M^6_{pair}	6-plane of a $\pm E$ pair	$\{Z : \mathrm{Re}\,V_2 \cdot Z = 0,\ \mathrm{Im}\,V_2 \cdot Z = 0\}$